



Network

University of Insubria

Security architecture

Security domain

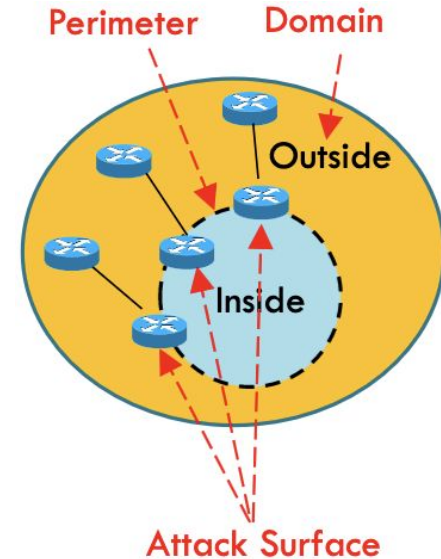
- a set of entities to be managed as a unique administration area

Security perimeter

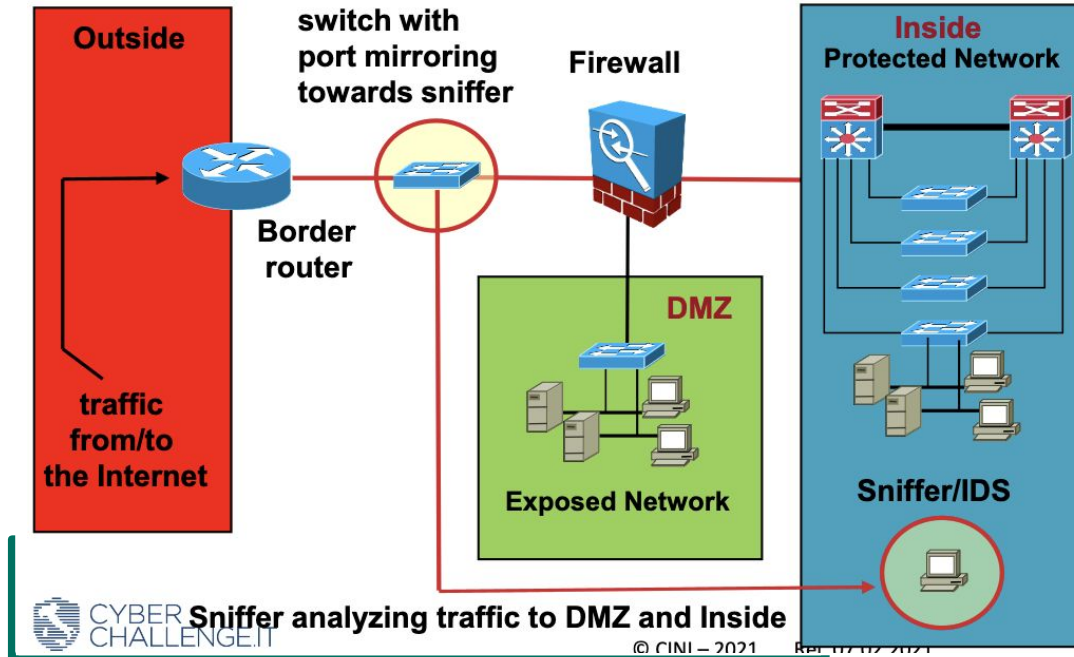
- secured boundary between the external and internal side of a security domain

Attack surface

- sum of the different points where an unauthorized entity can try to do any kind of unauthorized activity



Security architecture

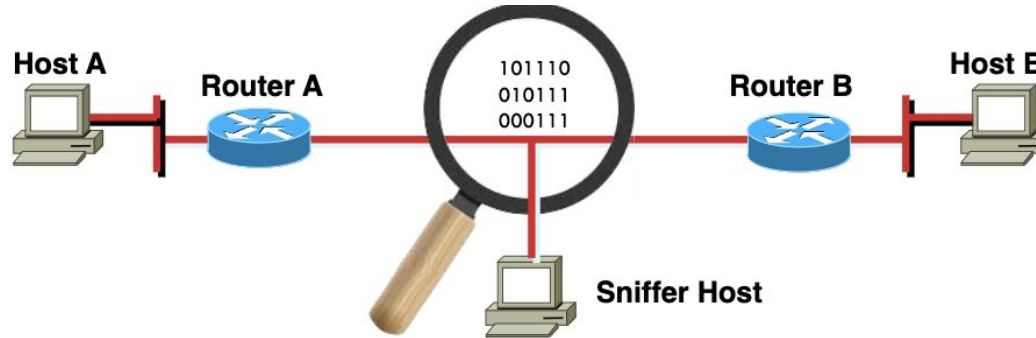


- DMZ (the set of internal machines that expose services outside)

Sniffing

A sniffer is a software application that is capable of acquiring packets at the datalink level

- It is able to interpret clear information relating to level 2, 3 and 4 packet headers as well as application level protocols such as: FTP, HTTP, etc.

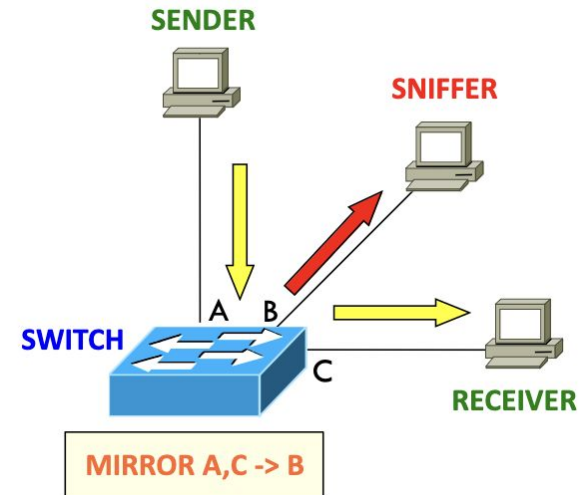


Sniffing on switched networks

Traffic is routed according to the MAC address + Port

- a sniffer is only able to intercept the traffic destined to its hosting

Idea: configure the switch port to which the sniffer is connected in mirroring mode



Other techniques

If port mirroring is not available

- Use repeater devices
- Use dedicated HW probes
- Perform traffic diversion through specific attacks (ARP Poisoning)

Tcpdump

Sniffer

- Software or hardware tool that by telling on promiscuous mode configuration captures and allows the analysis of all the packages that pass through a network segment

Tool

- tcpdump

Tcpdump filters

- Type: host, net e port
 - host 155.185.54.156
 - port 22
- Dir: src, dst, src or dst
 - src 155.185.54.156
- Proto: ether, fddi, tr, ip, ip6, arp, rarp, decnet, tcp and udp
 - tcp port 21
 - arp net 155.185.54

Tcpdump filters

- Expressions define the criteria with which to choose what has to be displayed
- Expressions consist of one or more primitives preceded by "qualifiers"

Source or destination host:	<code>host spoofed.target.net</code>
Destination network 172.31.x.x:	<code>dst net 172.31</code>
Destination networks 172.16 - 172.31:	<code>dst net 172 and (ip[17]>15) and (ip[17]<32)</code>
Source port 7:	<code>src port 7</code>
Destination port 19:	<code>dst port chargen</code>
Source port < 20:	<code>udp[0:2] < 20</code>
Destination port <20:	<code>udp[2:2] < 20</code>

Tcpdump

```
# tcpdump 'port 23'
```

```
10.6.1.9.4548 > 10.6.1.2.23: S 2115515278:2115515278(0) win 32120 <mss 1460,  
nop,nop,sackOK,nop,wscale 0> (DF)
```

```
10.6.1.2.23 > 10.6.1.9.4548: S 1220480853:1220480853(0)  
ack 2115515279 win 32120 <mss 1460,nop,nop,sackOK,nop,wscale 0> (DF)
```

```
10.6.1.9.4548 > 10.6.1.2.23: . ack 1220480854 win 32120 (DF)
```

Tcpdump output

- hosts can be referenced by name or IP address
- the ports can be specified by number or name of the service
- to specify a range of values, specific bytes must be pointed to

```
08:08:16.155 spoofed.target.net.7 > 172.31.203.17.chargen: udp
```

timestamp

src IP

src port

dst IP

dst port protocol

Network Traffic Analysis Via Wireshark

Wireshark

Wireshark is a tool to capture data from a network (sniffer) and to analyse them

- Analysis can be performed in real-time or on previously-recorded traffic files (PCAP files)

Wireshark & CTF

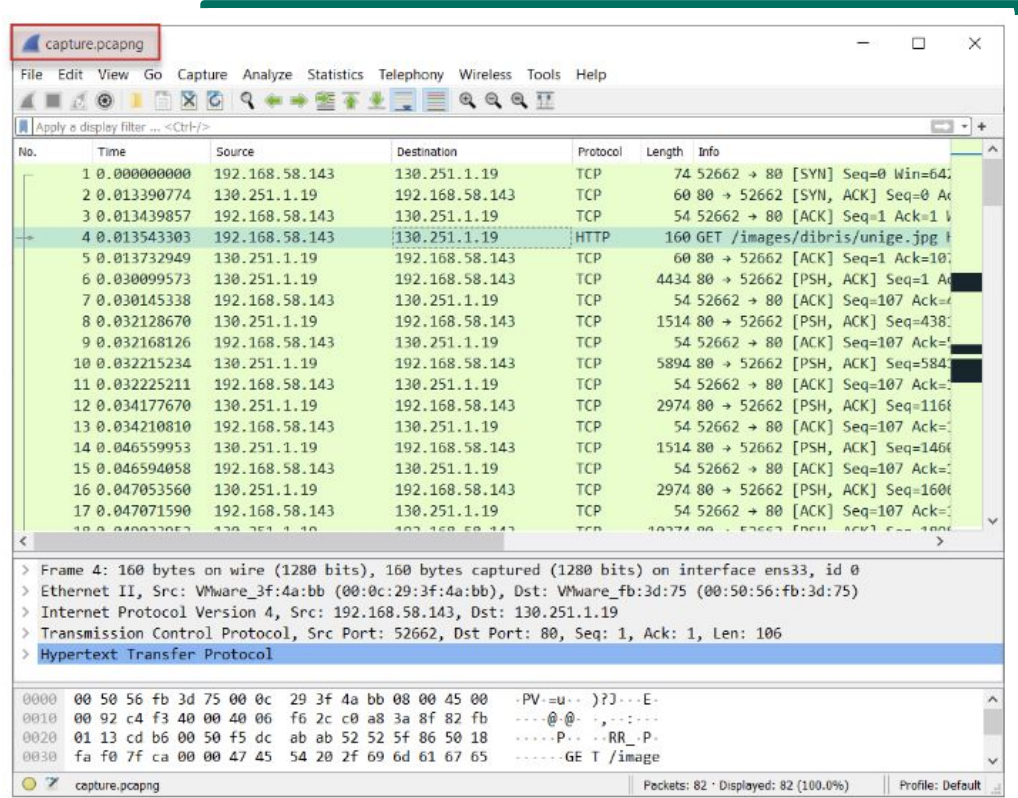
In some CTF challenges a PCAP file is given

- solving the challenges requires analyzing the capture to find the flag by
 - answering questions related to network traffic
 - carving file from packet streams

Wireshark

Wireshark provides a Graphical User Interface

- menu
- main toolbar
- filter toolbar
- packet list
- packet details
- packet bytes



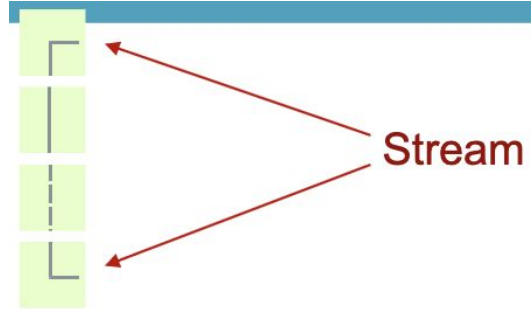
Wireshark

Packet list

No.	Time	Source	Destination	Protocol	Length	Info	columns
1	0.000000000	192.168.58.143	130.251.1.19	TCP	74	52662 → 80 [SYN] Seq=0 Win=64240 Len=0	
2	0.013390774	130.251.1.19	192.168.58.143	TCP	60	80 → 52662 [SYN, ACK] Seq=0 Ack=1 Win=6	
3	0.013439857	192.168.58.143	130.251.1.19	TCP	54	52662 → 80 [ACK] Seq=1 Ack=1 Win=64240	
4	0.013543303	192.168.58.143	130.251.1.19	HTTP	160	GET /images/dibris/unige.jpg HTTP/1.1	
5	0.013732949	130.251.1.19	192.168.58.143	TCP	60	80 → 52662 [ACK] Seq=1 Ack=107 Win=6424	
6	0.030099573	130.251.1.19	192.168.58.143	TCP	4434	80 → 52662 [PSH, ACK] Seq=1 Ack=107 Win	

Related packets symbols

- First packet in a conversation
- Part of the selected conversation
- Not part of the selected conversation
- Last packet in a conversation



- Request
- Response



- The selected packet acknowledges this packet
- The selected packet is a duplicate acknowledgement of this packet
- The selected packet is related to this packet in some other way



Challenge

Network 01 - Pcap

Network 02 - Dettagli

Network 01 - Pcap

nw-intro01.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	:::1	:::1	TCP	94	53422 → 9001 [SYN] Seq=0 Win=65476 Len=0 MSS=65476 SACK_PERM=1 TSval=615756801 TSecr=0 WS=128
2	0.000019625	:::1	:::1	TCP	74	9001 → 53422 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
3	0.000114816	127.0.0.1	127.0.0.1	TCP	74	58228 → 9001 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM=1 TSval=1831588898 TSecr=0 WS=128
4	0.000144864	127.0.0.1	127.0.0.1	TCP	74	9001 → 58228 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM=1 TSval=1831588898 TSecr=1831588898 WS=128
5	0.000166429	127.0.0.1	127.0.0.1	TCP	66	58228 → 9001 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=1831588898 TSecr=1831588898
6	71.149863898	127.0.0.1	127.0.0.1	TCP	96	58228 → 9001 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=30 TSval=1831660048 TSecr=1831588898
7	71.149912543	127.0.0.1	127.0.0.1	TCP	66	9001 → 58228 [ACK] Seq=1 Ack=31 Win=65536 Len=0 TSval=1831660048 TSecr=1831660048

Frame 6: 96 bytes on wire (768 bits), 96 bytes captured (768 bits) on interface lo, id 0

Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)

- Destination: 00:00:00:00:00:00 (00:00:00:00:00:00)
 -0. = LG bit: Globally unique address (factory default)
 -0. = IG bit: Individual address (unicast)
- Source: 00:00:00:00:00:00 (00:00:00:00:00:00)
 - Address: 00:00:00:00:00:00 (00:00:00:00:00:00)
 -0. = LG bit: Globally unique address (factory default)
 -0. = IG bit: Individual address (unicast)
- Type: IPv4 (0x0800)
- Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
 - 0100 = Version: 4
 - 0101 ... = Header Length: 20 bytes (5)

```

0000 00 00 00 00 00 00 00 00 00 00 00 08 00 45 00  ....E
0010 00 52 ee bb 40 00 40 06 4d e8 7f 00 00 01 7f 00  R.@.@.M....
0020 00 01 e3 74 23 29 89 e8 ba fe b2 85 6b c6 80 18  ...t#...k...
0030 02 00 fe 46 00 00 01 01 08 0a 6d 2c ea 10 6d 2b  ...F...m...m+
0040 d4 22 66 6c 61 67 7b 59 30 75 5f 6b 6e 30 77 5f  ...flag{Y 0u kn0w
0050 57 68 34 74 5f 61 5f 50 63 34 70 5f 31 73 7d 0a  Wh4t_a_P c4p_1s}
  
```

Network 02 - Dettagli

nw-intro02.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter Wireshark · Packet 4 · nw-intro0

No.	Time	
1	0.000000	
2	0.000091	
3	0.000141	
4	24.803856	
5	24.803946	
6	27.782942	
7	27.783189	
8	27.783315	

▼ Frame 4: 129 bytes
Encapsulation: t
Arrival Time: A
[Time shift for
Epoch Time: 164
[Time delta fro
[Time since ref
[Time since ref
Frame Number: 4
Frame Length: 1
Capture Length:
[Frame is marke
[Frame is ignor
[Protocols in f

Capture Length: 129 bytes (1032 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp:data]
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]

▼ Ethernet II, Src: 26:0e:ad:ef:c6:65 (26:0e:ad:ef:c6:65), Dst: a2:04:5b:f6:c1:09 (a2:04:5b:f6:c1:09)
▼ Destination: a2:04:5b:f6:c1:09 (a2:04:5b:f6:c1:09)
Address: a2:04:5b:f6:c1:09 (a2:04:5b:f6:c1:09)
...1. = LG bit: Locally administered address (this is NOT the factory default)
...0. = IG bit: Individual address (unicast)
▼ Source: 26:0e:ad:ef:c6:65 (26:0e:ad:ef:c6:65)
Address: 26:0e:ad:ef:c6:65 (26:0e:ad:ef:c6:65)
...1. = LG bit: Locally administered address (this is NOT the factory default)
...0. = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)

► Internet Protocol Version 4, Src: 192.168.100.25, Dst: 192.168.100.26
► Transmission Control Protocol, Src Port: 47850, Dst Port: 9001, Seq: 1, Ack: 1, Len: 63
▼ Data (63 bytes)
Data: 54686520666c616720697320696e2074686520666f726d617420666c61677b736f757263...
[Length: 63]

```

0000 a2 04 5b f6 c1 09 26 0e ad ef c6 65 08 00 45 00  ..[...&...e..E-
0010 00 73 ff f9 40 00 40 06 f1 06 c0 a8 64 19 c0 a8  -s_@@...d...
0020 64 1a ba ea 23 29 a3 49 66 bb 31 69 33 5d 80 18  d...#)I f li3]..
0030 01 f6 49 ea 00 01 01 08 0a bb 22 ab 96 ee 05  -I....."....
0040 27 f2 54 68 65 20 66 6c 61 67 20 69 73 20 69 6e  'The fl ag is in
0050 20 74 68 65 20 66 6f 72 6d 61 74 20 66 6c 61 67  the for mat flag
0060 7b 73 6f 75 72 63 65 5f 6d 61 63 5f 61 64 64 72  {source_mac_addr
0070 65 73 73 2f 64 61 74 61 5f 6c 65 6e 67 74 68 7d  ess/data _length}
0080 0a
  
```

Mark, ignore and comment

mark packets of particular interest

- CTRL-M
- jump forward SHIFT-CTRL-N and backward SHIFT-CTRL-B between marked packets

ignore packets

- CTRL-D

comment packets

- CTRL-ALT-C

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	145.254.160.237	65.208.228.223	TCP	62	3372 → 80 [SYN]
2	0.911310				62	<Ignored>
3	0.911310	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK]

marked ignored commented

Packet comments

Commented Comment content

Filtering packets

Wireshark provides a filter language to control which packets are displayed. Used to check for

- presence of a protocol or field
- value of a field
- compare two fields

Note: can be combined with logical operators and parentheses into complex expressions



The image shows a Wireshark packet capture. On the left, the packet details pane shows an IP packet with the following fields: Header Checksum: 0x91ee [validation disabled], [Header checksum status: Unverified], Source Address: 145.254.160.237, Destination Address: 65.208.228.223, and Transmission Control Protocol, Seq: 3372, Dest Port: 80. A red box highlights the 'Source Address' field, and a red arrow points from it to the filter bar on the right. The filter bar contains the expression 'ip.src == 145.254.160.237'. Below the filter bar, the packet list pane shows a table with columns 'No.', 'Time', and 'Source'.

Protocol field

Prepare a Filter

Header Checksum: 0x91ee [validation disabled]
[Header checksum status: Unverified]
Source Address: 145.254.160.237
Destination Address: 65.208.228.223
Transmission Control Protocol, Seq: 3372, Dest Port: 80

ip.src == 145.254.160.237

No.	Time	Source
-----	------	--------

Filtering packets

Comparison operators (Equal ==, Not Equal !=, Greater Than >, Less Than <, ..)

- `http.request.method == "POST"`

All, Any

- `all tcp.port > 1024, any ip.addr != 1.1.1.1`

contains, matches

- `http contains "https://www.wireshark.org", wsp.header.user_agent matches "(?-i)cldc"`

Coloring rules

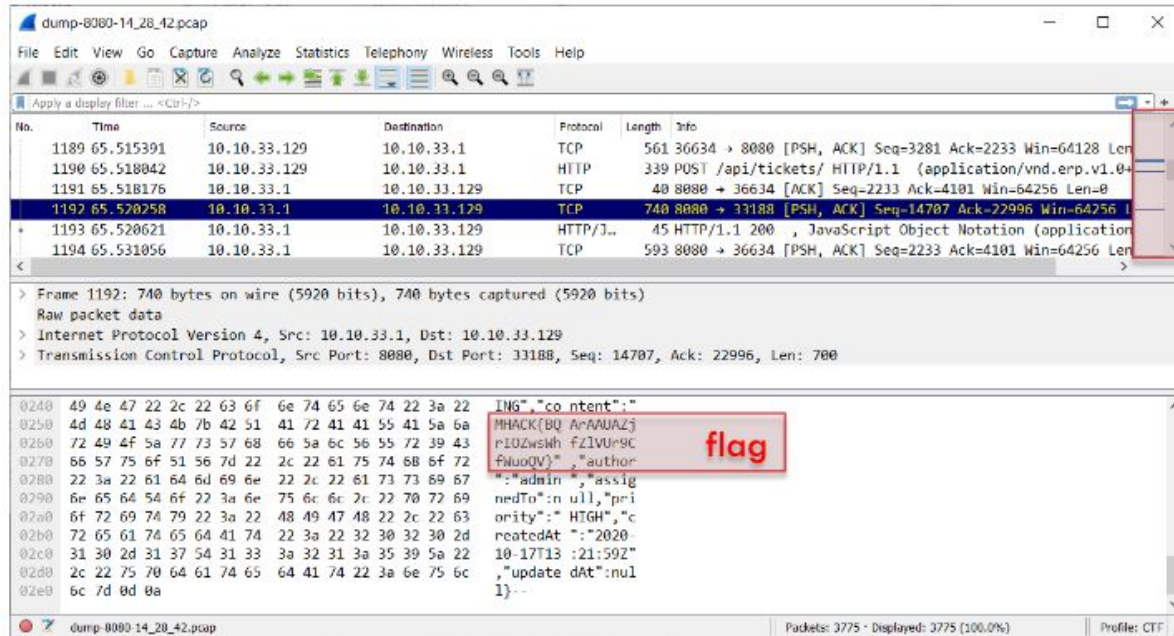
Wireshark supports coloring rules for packets

- View → Coloring Rules

Example

- Get the flag format: `CCIT\{[A-Za-z0-9-_{32}\}`
- View → Coloring Rules...
- Add a new rule for highlighting flags using the “match” keyword

Coloring rules



Packet
with flags
(blue lines)

flag

Challenge

Network 03 - Filtri 1

Network 04 - Filtri 2

Network 05 - Filtri 3

Network 06 - Filtri 4

Network 03 - Filtri 1

nw-int

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
176	320.391854278	192.168.100.1	35.190.56.182	HTTP	188	GET / HTTP/1.1
178	320.554032265	35.190.56.182	192.168.100.1	HTTP	361	HTTP/1.1 301 Moved Permanently (text/html)

▼ Frame 176: 188 bytes on wire (1504 bits), 188 bytes captured (1504 bits) on interface veth0, id 0

Interface id: 0 (veth0)

Encapsulation type: Ethernet (1)

Arrival Time: Apr 15, 2022 13:03:12.871394694 CEST

[Time shift for this packet: 0.000000000 seconds]

Epoch Time: 1650020592.871394694 seconds

[Time delta from previous captured frame: 0.000234582 seconds]

[Time delta from previous displayed frame: 0.000000000 seconds]

[Time since reference or first frame: 320.391854278 seconds]

Frame Number: 176

Frame Length: 188 bytes (1504 bits)

Capture Length: 188 bytes (1504 bits)

[Frame is marked: False]

[Frame is ignored: False]

```

0000 b6 2b e7 17 82 7a a6 80 1c 3d c4 bf 08 00 45 00  -+...z...=...E-
0010 00 ae f9 c0 40 00 40 06 bf 6b c0 a8 64 01 23 be  -...@...k..d#-
0020 38 b6 e2 62 00 50 c0 0e 2d e7 33 fa e1 74 80 18  8..b.P...-3..t-
0030 01 f6 81 be 00 00 01 01 08 0a 8c 90 d2 04 d7 57  -.....W
0040 01 87 47 45 54 20 2f 20 48 54 54 50 2f 31 2e 31  -GET / HTTP/1.1
0050 0d 0a 48 6f 73 74 3a 20 70 77 6e 2e 66 6c 61 67  -Host: pwn.flag
0060 2e 6f 72 67 0d 0a 55 73 65 72 2d 41 67 65 6e 74  -..org..Us er-Agent
0070 3a 20 63 75 72 6c 2f 37 2e 36 34 2e 30 0d 0a 41  -: curl/7.64.0..A
0080 63 63 65 70 74 3a 20 2a 2f 2a 0d 0a 46 6c 61 67  -cept: */*..flag
0090 2d 53 74 72 69 6e 67 3a 20 66 6c 61 67 7b 4c 33  -String: flag{L3
00a0 61 52 6e 31 4e 39 5f 34 62 30 75 54 5f 46 31 6c  -aRn1N9 4 b0uT_Fil
00b0 74 65 72 35 5f 70 31 7d 0d 0a 0d 0a  -ter5_p1} ----
  
```

Network 04 - Filtri 2

nw-intro03.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ip.src == 192.168.100.3 && dns

No.	Time	Source	Destination	Protocol	Length	Info
292	683.065614141	192.168.100.3	1.1.1.1	DNS	102	Standard query 0xe516 A flag{L3aRn1N9 4b0uT Filter5 1P DNS f1lt3r}

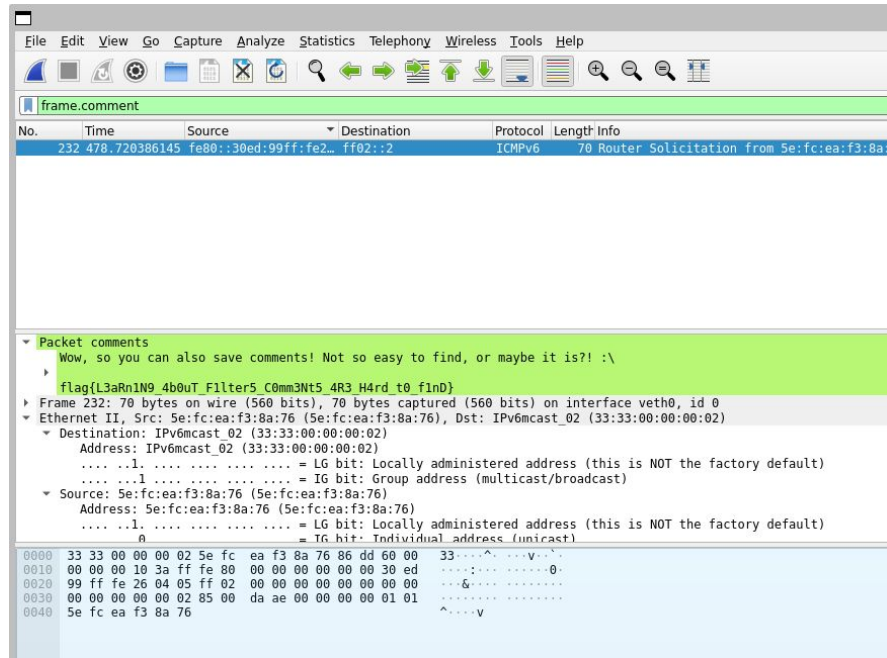
Domain Name System (query)

- Transaction ID: 0xe516
- Flags: 0x0100 Standard query
- Questions: 1
- Answer RRs: 0
- Authority RRs: 0
- Additional RRs: 0
- Queries
 - flag{L3aRn1N9 4b0uT Filter5 1P DNS f1lt3r}: type A, class IN
 - Name: flag{L3aRn1N9 4b0uT Filter5 1P DNS f1lt3r}
 - [Name Length: 42]
 - [Label Count: 1]
 - Type: A (Host Address) (1)
 - Class: IN (0x0001)

```

0000 b6 2b e7 17 82 7a be 01 90 55 5e 4a 08 00 45 00  .+...z...U^]..E.
0010 00 58 f6 4e 00 00 40 11 5d 99 c0 a8 64 03 01 01  .X.N-@. ]...d...
0020 01 01 d0 2e 00 35 00 44 27 03 e5 16 01 00 00 01  ...5.D .....
0030 00 00 00 00 00 00 2a 65 6c 61 67 7b 4c 33 61 52  ....*flag{L3aR
0040 6e 51 4e 39 5f 34 62 30 75 54 5f 40 31 6c 74 65  n1N9 4b0 uT flt3r
0050 72 35 5f 31 50 5f 44 4e 35 5f 66 31 6c 74 33 72  r5 1P DN 5 f1lt3r
0060 7d 00 00 01 00 01  }.....
  
```

Network 05 - Filtri 3



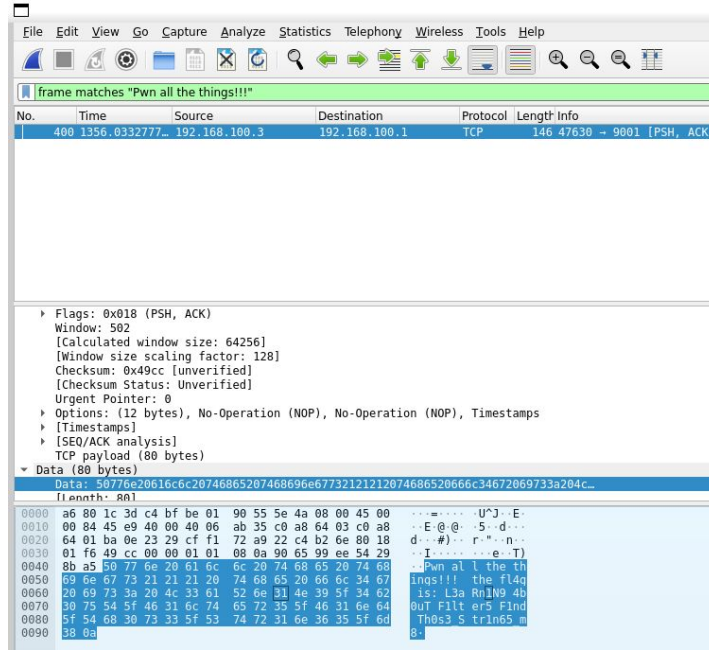
The image shows a Wireshark packet capture analysis of an ICMPv6 Router Solicitation packet. The packet list pane shows a single packet (No. 232) at time 478.720386145, sourced from fe80::30ed:99ff:fe2... and destined to ff02::2. The protocol is ICMPv6, and the length is 70 bytes. The packet information pane shows the packet details, including the Ethernet II header and the ICMPv6 Router Solicitation message. The packet bytes pane shows the raw data of the packet, including the Ethernet II header and the ICMPv6 Router Solicitation message.

No.	Time	Source	Destination	Protocol	Length	Info
232	478.720386145	fe80::30ed:99ff:fe2...	ff02::2	ICMPv6	70	Router Solicitation from 5e:fc:ea:f3:8a:

Packet details:
 Ethernet II, Src: 5e:fc:ea:f3:8a:76 (5e:fc:ea:f3:8a:76), Dst: IPv6mcast_02 (33:33:00:00:00:02)
 Destination: IPv6mcast_02 (33:33:00:00:00:02)
 Address: IPv6mcast_02 (33:33:00:00:00:02)
 ...1. = LG bit: Locally administered address (this is NOT the factory default)
 ...1. = IG bit: Group address (multicast/broadcast)
 Source: 5e:fc:ea:f3:8a:76 (5e:fc:ea:f3:8a:76)
 Address: 5e:fc:ea:f3:8a:76 (5e:fc:ea:f3:8a:76)
 ...1. = LG bit: Locally administered address (this is NOT the factory default)
 ...1. = IG bit: Individual address (unicast)

Packet bytes:
 0000 33 33 00 00 00 02 5e fc ea f3 8a 76 86 dd 60 00 33...^...v...
 0010 00 00 00 10 3a ff fe 00 00 00 00 00 00 30 ed ...:...:0...
 0020 99 ff fe 26 04 05 ff 02 00 00 00 00 00 00 00 ...&...:...
 0030 00 00 00 00 00 02 85 00 da ae 00 00 00 01 01 ...:...:...
 0040 5e fc ea f3 8a 76 ^...v

Network 06 - Filtri 4



Wireshark interface showing a network packet capture. The packet list displays a single packet (No. 400) at Time 1356.0332777, Source 192.168.100.3, Destination 192.168.100.1, Protocol TCP, Length 146. The packet details pane shows the following information:

- Flags: 0x018 (PSH, ACK)
- Window: 502
- [Calculated window size: 64256]
- [Window size scaling factor: 128]
- Checksum: 0x49cc [unverified]
- [Checksum Status: Unverified]
- Urgent Pointer: 0
- Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
- [Timestamps]
- [SEQ/ACK analysis]
- TCP payload (80 bytes)

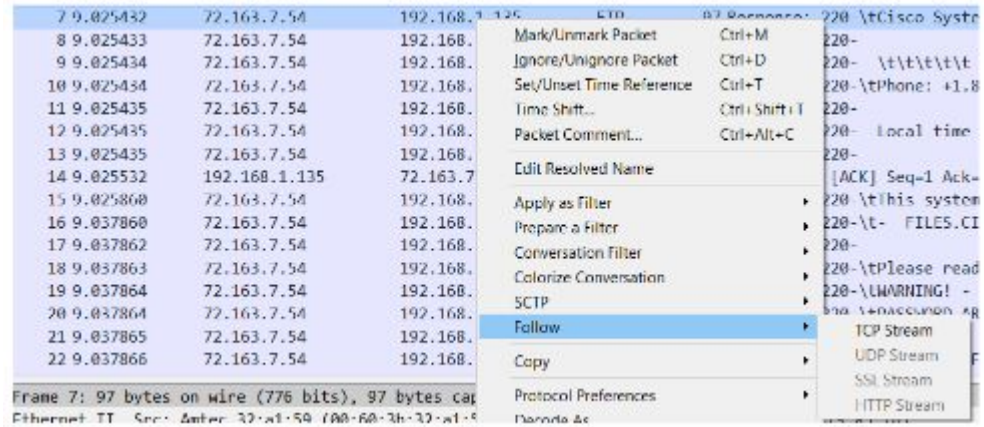
The packet data (80 bytes) is displayed in hexadecimal and ASCII format:

```
0000 a6 80 1c 3d c4 bf be 01 90 55 5e 4a 08 00 45 00  ...-...UJ E
0010 00 84 45 e9 40 00 00 06 ab 35 c0 a8 64 03 c0 a8  ..E@.5.d...
0020 64 01 ba 0e 23 29 cf f1 72 a9 22 c4 b2 6e 80 18  d..#)..r"...n..
0030 01 f6 49 cc 00 00 01 01 08 0a 90 65 99 ee 54 29  ..I.....e.T)
0040 8b a5 f0 77 6e 20 61 6c 6c 20 74 68 65 20 74 68  ..Pwn al l the th
0050 69 0e 67 73 21 21 21 20 74 68 65 20 66 6c 34 67  ings!!! the flds
0060 28 69 73 3a 20 4c 33 61 52 6e 31 4e 39 5f 34 62  1s: L3a RnIn0 4b
0070 30 75 54 5f 46 31 6c 74 65 72 35 5f 46 31 6e 64  0uT Filt or5 FInd
0080 5f 54 68 30 73 33 5f 53 74 72 31 6e 36 35 5f 6d  Th0s3 $ trIn65 m
0090 38 0a                                     8.
```

Follow streams

Instead of individual packets, it is possible to see data flowing between client and server

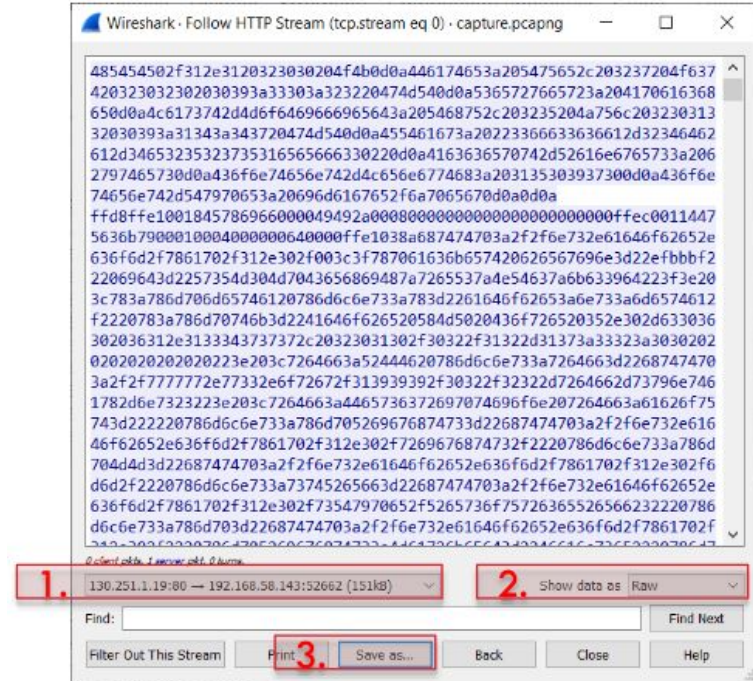
- extract and save a JPEG file downloaded using HTTP
- extract a file from a generic stream



Follow streams

Extract and save a JPEG file downloaded using HTTP

- Select server → client packets (Blue)
- Show data as Raw
- Save as...
- Remove the HTTP header from the saved file



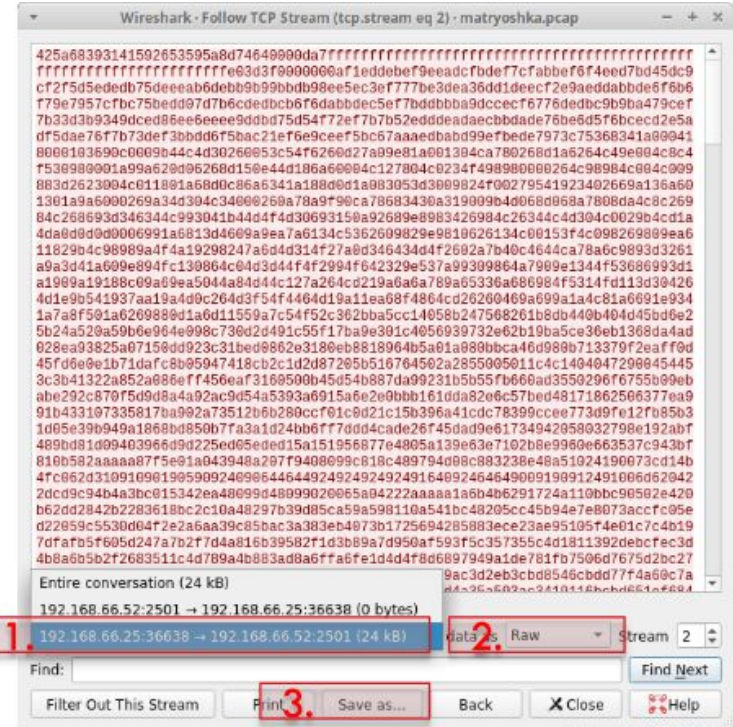
Follow streams

Extract a file from a generic stream

- Switch between client to server or server to client conversation
- Show data as Raw
- Save as... (e.g., /tmp/bin1)
- Use the linux file utility to determine the file type

```

/tmp file bin1
bin1: bzip2 compressed data, block size = 900k
  
```



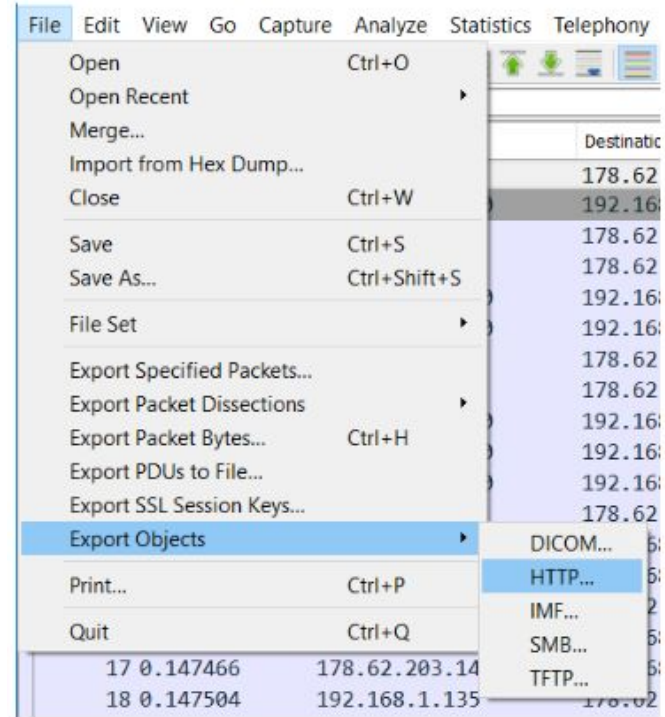
Follow streams

Export ...

- File -> Export ...

Export objects

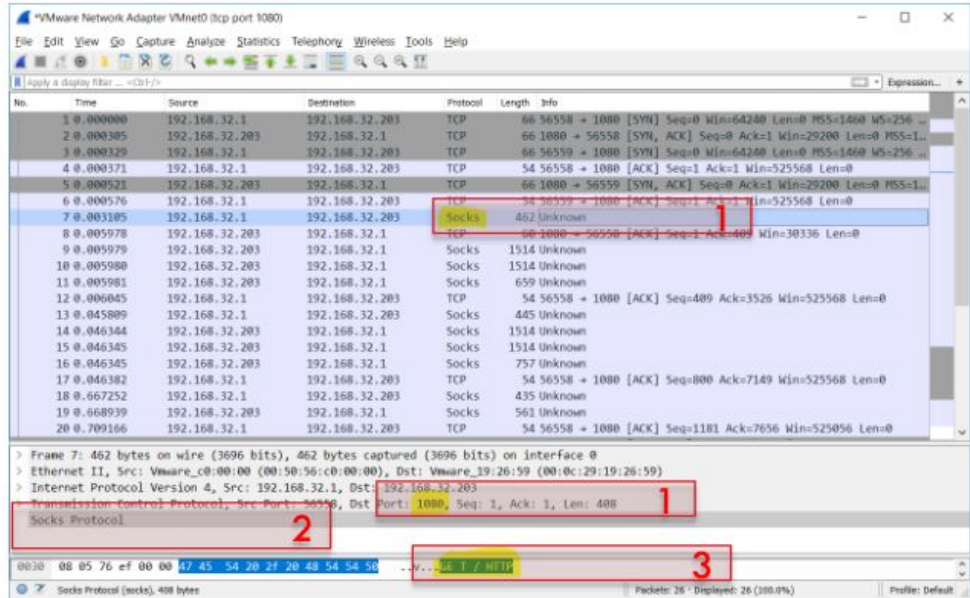
- File → Export Objects
- scans through the protocol streams and takes reassembled objects



Dissectors

Dissectors are what parse a protocol and decode it for presenting on the interface

- each protocol has its own dissector
- find the right dissector to start decoding the packet data

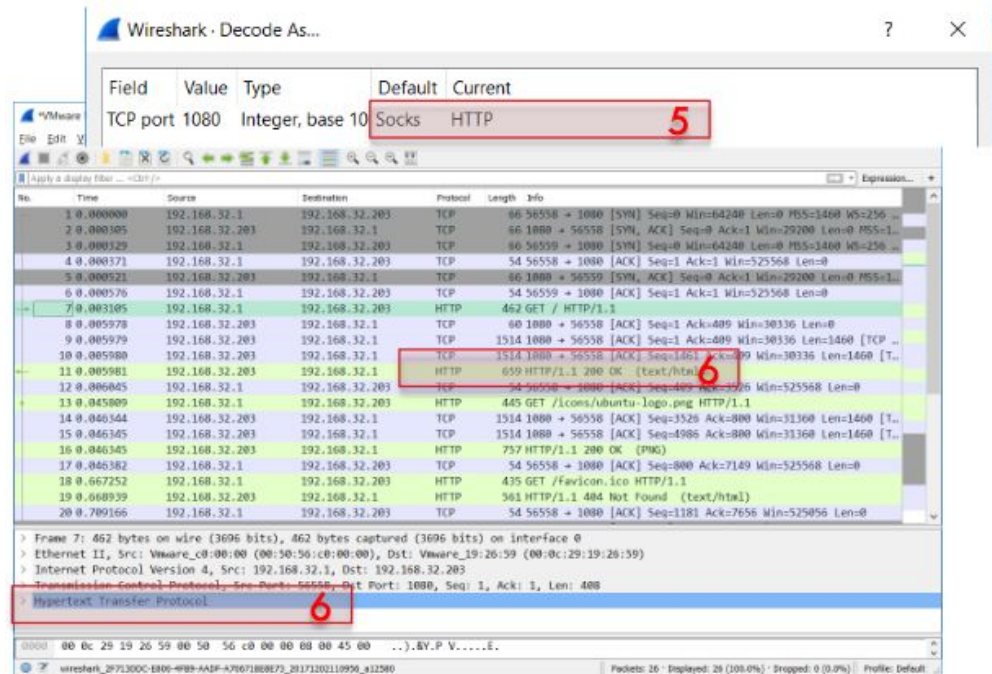


The screenshot displays the Wireshark network protocol analyzer interface. The top pane shows a list of captured packets. Packet 7 is selected, which is a Socks protocol packet. The middle pane shows the details of this packet, including the Socks protocol structure with fields like 'Socks Protocol', 'Port', and 'Seq'. The bottom pane shows the raw data of the packet in hexadecimal and ASCII. Red boxes and numbers 1, 2, and 3 are used to highlight specific parts of the interface: 1 points to the 'Socks Protocol' field in the details pane, 2 points to the 'Socks Protocol' field in the packet list pane, and 3 points to the 'Socks Protocol (socks), 408 bytes' field in the packet bytes pane.

Dissectors

Example

- wireshark applies a Socks dissector, as the well-known port for Socks traffic is 1080/tcp
- the dissector is not able to decode the data correctly
- change the Current value (Socks) with the right dissector (HTTP)



Challenge

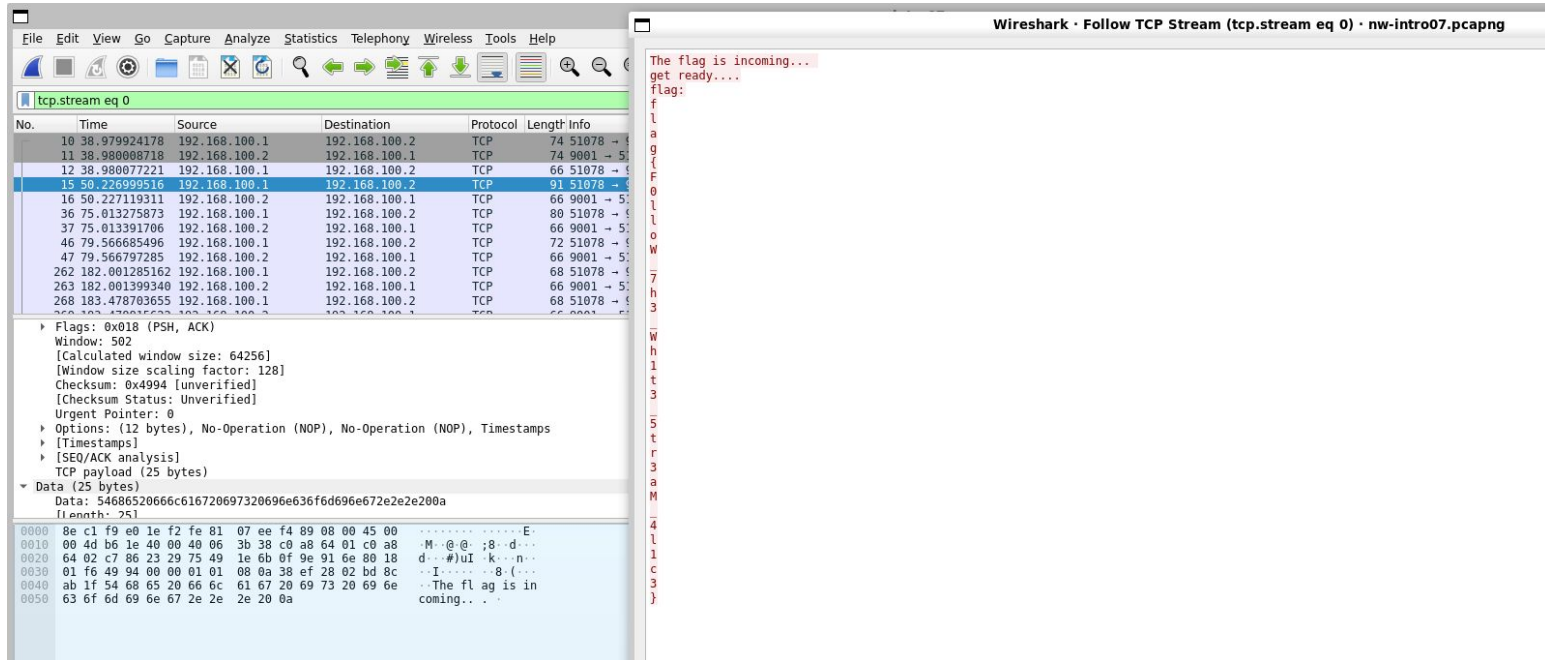
Network 07 - Stream TCP

Network 08 - Estrazione di bytes

Network 09 - Decifrare traffico TLS

Network 10 - Esporta oggetti

Network 07 - Stream TCP



The image displays the Wireshark network protocol analyzer interface. The main pane shows a list of network packets, with the selected packet (No. 15) highlighted. The packet details pane on the right shows the structure of the selected TCP packet, including flags, window size, and payload. The packet list pane on the left shows the sequence of packets in the stream.

Wireshark - Follow TCP Stream (tcp.stream eq 0) - nw-intro07.pcapng

No.	Time	Source	Destination	Protocol	Length	Info
10	38.979924178	192.168.100.1	192.168.100.2	TCP	74	51078 → 51078
11	38.980008718	192.168.100.2	192.168.100.1	TCP	74	51078 → 51078
12	38.980077221	192.168.100.1	192.168.100.2	TCP	66	51078 → 51078
15	50.226999516	192.168.100.1	192.168.100.2	TCP	91	51078 → 51078
16	50.227119311	192.168.100.2	192.168.100.1	TCP	66	9001 → 51078
36	75.013275873	192.168.100.1	192.168.100.2	TCP	80	51078 → 51078
37	75.013391706	192.168.100.2	192.168.100.1	TCP	66	9001 → 51078
46	79.566685496	192.168.100.1	192.168.100.2	TCP	72	51078 → 51078
47	79.566797285	192.168.100.2	192.168.100.1	TCP	66	9001 → 51078
262	182.001285162	192.168.100.1	192.168.100.2	TCP	68	51078 → 51078
263	182.001399340	192.168.100.2	192.168.100.1	TCP	66	9001 → 51078
268	183.478703655	192.168.100.1	192.168.100.2	TCP	68	51078 → 51078

Packet 15 Details:

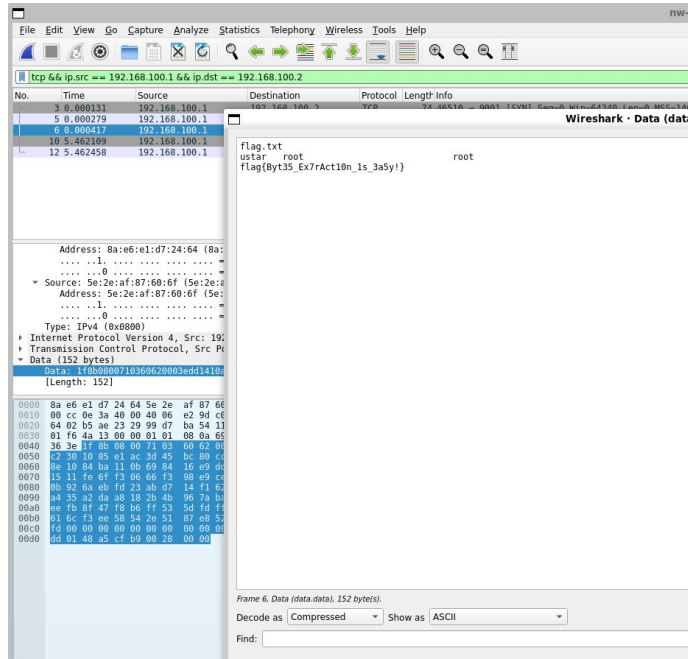
- Flags: 0x018 (PSH, ACK)
- Window: 502
- [Calculated window size: 64256]
- [Window size scaling factor: 128]
- Checksum: 0x4994 [unverified]
- [Checksum Status: Unverified]
- Urgent Pointer: 0
- Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
- [Timestamps]
- [SEQ/ACK analysis]
- TCP payload (25 bytes)
- Data (25 bytes)
- Data: 5468652066c616720697320696e636f6d696e672e2e200a
- Length: 251

Hex/ASCII View:

```

0000  8e c1 f9 e0 1e f2 fe 81 07 ee f4 89 08 00 45 00  ....E..
0010  00 4d b6 1e 40 00 40 06 3b 38 c0 a8 64 01 c0 a8  .M..@. ;8..d..
0020  64 02 c7 86 23 29 75 49 1e 6b 0f 9e 91 6e 80 18  d...#)U!-k...n..
0030  91 f6 49 94 02 00 01 01 08 0a 38 ef 28 02 bd 8c  .I...8 (...
0040  ab 1f 54 68 65 20 66 6c 61 67 20 69 73 20 69 6e  .The fl ag is in
0050  63 6f 6d 69 6e 67 2e 2e 2e 20 0a                coming..
  
```

Network 08 - Estrazione di bytes



The image shows a Wireshark network packet capture analysis. The top pane displays a list of captured packets. The second pane shows the details of the selected packet (No. 6), including the Ethernet II header, Internet Protocol Version 4 header, and the Data (152 bytes) field. The third pane shows the raw data of the selected packet in hexadecimal and ASCII format.

Filter: `tcp && ip.src == 192.168.100.1 && ip.dst == 192.168.100.2`

No.	Time	Source	Destination	Protocol	Length	Info
3	0.000131	192.168.100.1	192.168.100.2	ICMP	64	Echo (ping) request
5	0.000279	192.168.100.1	192.168.100.2	ICMP	64	Echo (ping) request
6	0.000417	192.168.100.1	192.168.100.2	TCP	60	65535 → 65535 [RST] Seq=103456789 Win=0 Len=0
10	5.462109	192.168.100.1	192.168.100.2	TCP	60	65535 → 65535 [RST] Seq=103456789 Win=0 Len=0
12	5.462458	192.168.100.1	192.168.100.2	TCP	60	65535 → 65535 [RST] Seq=103456789 Win=0 Len=0

Wireshark · Data (data).

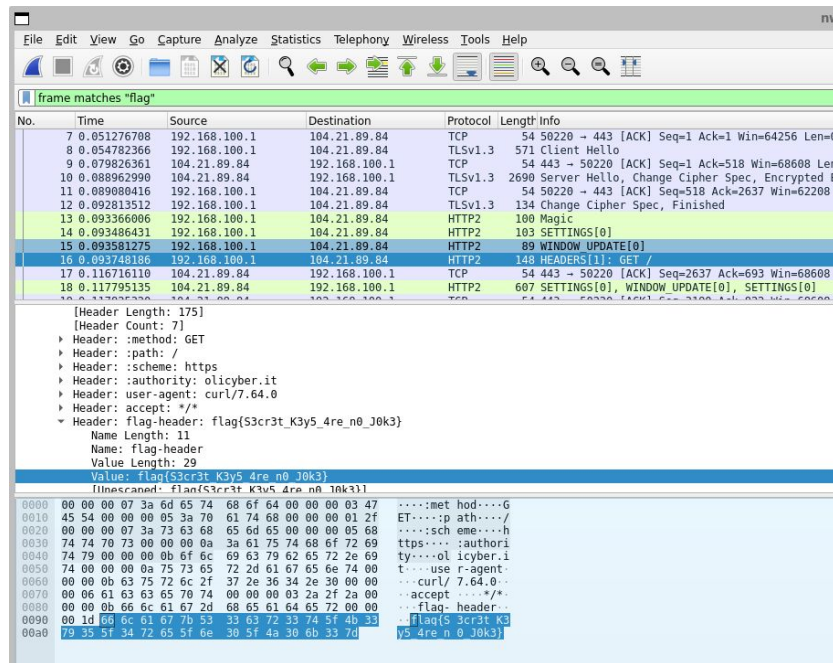
flag.txt
ustar root
flag(Byte35_Ex7rAct10n_1s_3a5y!)

Address: 8a:e6:e1:d7:24:64 (8a:...)
... ..
... ..
Source: 5e:2e:af:87:60:6f (5e:2e:af:87:60:6f)
Address: 5e:2e:af:87:60:6f (5e:2e:af:87:60:6f)
... ..
... ..
Type: IPv4 (0x0800)
Internet Protocol Version 4, Src: 192.168.100.1, Dst: 192.168.100.2
Transmission Control Protocol, Src Port: 65535, Dst Port: 65535
Data (152 bytes)
Data: 1f0b000071036062003edd1410... (Length: 152)

0000 8a e6 e1 d7 24 64 5e 2e af 87 66
0010 00 cc 0e 3a 48 00 48 06 e2 9d c4
0020 64 02 b5 ae 23 99 07 b3 54 11
0030 01 f6 4a 13 00 00 01 01 08 0a 66
0040 36 3e 1f 0b 00 00 71 03 60 62 00
0050 00 00 00 00 00 00 00 00 00 00 00
0060 0e 10 84 ba 11 00 69 84 16 e9 d6
0070 15 11 4e d7 13 00 66 13 98 e9 c4
0080 0b 92 6a eb fd 23 ab d7 14 f1 62
0090 a4 35 a2 da a8 18 2b 4b 96 7a b3
00a0 ee fb 87 47 f0 b6 ff 52 5d fd f1
00b0 e1 6c f3 ee 58 54 2e 51 87 e8 52
00c0 fd 00 00 00 00 00 00 00 00 00 00
00d0 fd 01 4b 35 c7 19 00 28 00 00 00

Frame 6: Data (data.data), 152 byte(s).
Decode as: Compressed Show as: ASCII
Find:

Network 09 - Decifrare traffico TLS



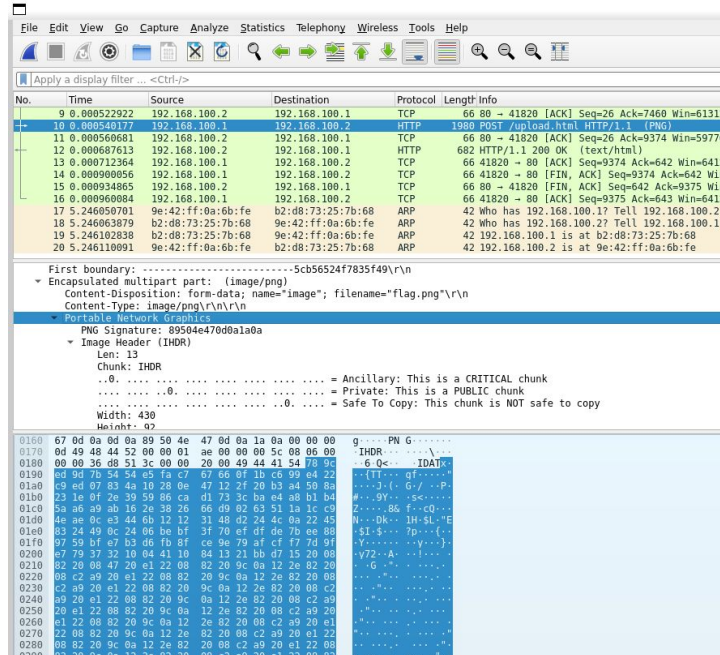
frame matches "flag"

No.	Time	Source	Destination	Protocol	Length	Info
7	0.051276788	192.168.100.1	104.21.89.84	TCP	54	50220 → 443 [ACK] Seq=1 Ack=1 Win=64256 Len=0
8	0.054782366	192.168.100.1	104.21.89.84	TLSv1.3	571	Client Hello
9	0.079826361	104.21.89.84	192.168.100.1	TCP	54	443 → 50220 [ACK] Seq=1 Ack=518 Win=68608 Len=0
10	0.089962990	104.21.89.84	192.168.100.1	TLSv1.3	2690	Server Hello, Change Cipher Spec, Encrypted E
11	0.089080416	192.168.100.1	104.21.89.84	TCP	54	50220 → 443 [ACK] Seq=518 Ack=2637 Win=62208 Len=0
12	0.092813512	192.168.100.1	104.21.89.84	TLSv1.3	134	Change Cipher Spec, Finished
13	0.093366006	192.168.100.1	104.21.89.84	HTTP2	100	Magic
14	0.093486431	192.168.100.1	104.21.89.84	HTTP2	103	SETTINGS[0]
15	0.093581275	192.168.100.1	104.21.89.84	HTTP2	89	WINDOW_UPDATE[0]
16	0.093748186	192.168.100.1	104.21.89.84	HTTP2	148	HEADERS[1]: GET /
17	0.116716110	104.21.89.84	192.168.100.1	TCP	54	443 → 50220 [ACK] Seq=2637 Ack=693 Win=68608 Len=0
18	0.117795135	104.21.89.84	192.168.100.1	HTTP2	607	SETTINGS[0], WINDOW_UPDATE[0], SETTINGS[0]

[Header Length: 175]
[Header Count: 7]
 ▶ Header: :method: GET
 ▶ Header: :path: /
 ▶ Header: :scheme: https
 ▶ Header: :authority: olicyber.it
 ▶ Header: user-agent: curl/7.64.0
 ▶ Header: accept: /*/*
 ▶ Header: flag-header: flag{53cr3t_K3y5_4re_n0_J0k3}
 Name Length: 11
 Name: flag-header
 Value Length: 29
 Value: flag{53cr3t_K3y5_4re_n0_J0k3}
 [linescaned: flag{53cr3t_K3y5_4re_n0_J0k3}]

```

0000  00 00 00 07 3a 6d 65 74 68 6f 64 00 00 00 03 47  ....met hod....G
0010  45 54 00 00 00 05 3a 70 61 74 68 00 00 00 01 2f  ET....p ath..../
0020  00 00 00 07 3a 73 63 68 65 6d 65 00 00 00 05 68  ....sch eme....h
0030  74 74 70 73 00 00 00 0a 3a 61 75 74 68 6f 72 69  ttps...:authori
0040  74 79 00 00 00 0b 6f 6c 69 63 79 62 65 72 2e 69  ty....ol icyber.i
0050  74 00 00 00 0a 75 73 65 72 2d 61 67 65 6e 74 00  t...use r-agent-
0060  00 00 0b 63 75 72 6c 2f 37 2e 36 34 2e 30 00 00  ...curl/ 7.64.0..
0070  00 06 61 63 63 65 70 74 00 00 00 03 2a 2f 2a 00  ...accept ....*/.*
0080  00 00 0b 6c 61 67 2d 68 65 61 64 65 72 00 00 00  ...flag-header-
0090  00 1d 68 6c 61 67 2d 68 65 61 64 65 72 00 00 00  ...flag{53cr3t_K3
00a0  79 35 5f 34 72 65 5f 6e 30 5f 4a 30 6b 33 7d  ...y5_4re_n0_J0k3}
  
```

Challenges

NS_1.01 - Sanity Check

NS_1.02 - No comment

NS_1.03 - Well-hidden ports

NS_1.04 - We are the robots

NS_1.05 - Name Swap

NS_1.06 - A naughty attacker

NS_1.07 - Dreamliner

Challenges

NS_2.01 - Secret web server

NS_2.02 - Nano nano

NS_2.03 - CCIT Unnecessary censorship

NS_2.04 - Gamers secret

NS_2.05 - Dynamic web yoga

NS_2.06 - IPV-4